

Synopsis

**K-Means Clustering–Based Analytical System for
Agricultural Region Classification.**

➤ **Problem ID: 04**

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- **Problem Description:**

- Agriculture is one of the most data-rich sectors, yet many regional planning bodies lack the tools to systematically classify agricultural zones based on multi-dimensional environmental and productivity indicators. This project addresses that gap by designing an intelligent analytical system that applies K-Means clustering to group agricultural regions into meaningful clusters based on 20 region-level features spanning environmental conditions, soil health, farming practices, and crop productivity metrics.
- The core challenge is that agricultural regions vary enormously across rainfall, temperature, soil composition, irrigation access, and yield outputs. Identifying natural groupings among these regions allows policymakers, agronomists, and agricultural scientists to tailor interventions, optimize resource allocation, and benchmark performance across comparable zones. Rather than relying on administrative boundaries or expert intuition alone, this system leverages unsupervised machine learning to discover data-driven groupings that reflect genuine similarities in agricultural profiles.
- The system must handle the full analytical pipeline — from raw data ingestion and exploratory analysis, through preprocessing and feature selection, to clustering, evaluation, and prediction on unseen regional data points.

- **Steps of Implementation:**

- **Dataset Construction and Loading** A structured synthetic or real-world dataset of agricultural regions is created or loaded, containing 20 features per region: average rainfall (mm), average temperature (°C), soil moisture level (%), soil pH, nitrogen content, phosphorus content, potassium content, irrigation coverage (%), crop area (hectares), fertilizer usage (kg/hectare), pesticide usage (kg/hectare), sunlight hours, humidity (%), altitude (meters), wind speed (km/h), groundwater level (meters), crop diversity index, farming mechanization score, historical yield variance, and average crop yield (tons/hectare). Each row represents one agricultural region.
- **Exploratory Data Analysis (EDA)** The dataset is thoroughly explored to understand the distribution, spread, and relationships among features. Summary statistics are computed for all 20 features. Univariate distributions are examined using histograms and box plots to detect skewness and outliers. A correlation heatmap is generated to identify redundant or highly correlated feature pairs. Pairplots or scatter matrix plots are used to examine joint distributions across key features.

Data Preprocessing Missing values, if present, are handled through median imputation for numerical features to preserve distributional properties. Outliers are identified using the IQR method and either capped or treated appropriately. Since K-Means is a distance-based algorithm, all features are scaled using StandardScaler (zero mean, unit variance) or MinMaxScaler to ensure no single feature dominates the distance computation due to its magnitude.

- **Feature Selection via Forward or Backward Elimination** To reduce noise and dimensionality while retaining meaningful signals, feature elimination is applied. Backward elimination begins with all 20 features and iteratively removes the least informative features based on criteria such as low variance, high collinearity (using VIF — Variance Inflation Factor), or low contribution to clustering quality as measured by Silhouette Score. Alternatively, forward selection starts with the most impactful feature and adds features incrementally. The final selected feature subset is used for all subsequent steps.
- **Prediction on New Unseen Data** The trained K-Means model is used to predict the cluster label for one or more new, unseen agricultural regions. The new data point is passed through the same preprocessing and feature selection pipeline (using the already-fitted scaler and selected feature columns), and the model's predict() method assigns it to the nearest cluster centroid. The assigned cluster and its characteristic profile are reported, demonstrating the system's practical deployability.
- **Expected Modules/ Libraries to be used :**
 - NumPy is used for numerical operations, array manipulation, and random data generation. Pandas is the primary tool for dataset creation, loading, manipulation, missing value handling, and feature engineering.
 - **Machine Learning** Scikit-learn provides the complete ML infrastructure: KMeans for the clustering algorithm, StandardScaler and MinMaxScaler for feature normalization, PCA for dimensionality reduction, silhouette_score and silhouette_samples for evaluation, and SimpleImputer for missing value treatment. The variance_inflation_factor function from statsmodels.stats.outliers_influence is used during feature selection to detect multicollinearity.
 - **Visualization** Matplotlib is the foundational plotting library used for histograms, elbow plots, silhouette plots, and custom charts. Seaborn is used on top of Matplotlib to generate aesthetically refined correlation heatmaps, pairplots, box plots, and cluster profile bar charts. Plotly (optional) can be used for interactive 3D cluster visualizations.
 - **Dimensionality Reduction** Scikit-learn's PCA is the primary tool for reducing dimensions before and after clustering. TSNE from scikit-learn can be used as an alternative or complement to PCA for non-linear structure detection in higher-dimensional data.
 - **Utilities** Warnings and os modules are used for suppressing non-critical warnings and managing file paths. Joblib or Pickle can be used to serialize the trained KMeans model and scaler for future inference.

- **Expected Evaluation metrics and Plots to be used:**

Evaluation Metrics

The Silhouette Score (overall) is the primary evaluation metric, measuring the mean ratio of intra-cluster cohesion to inter-cluster separation across all data points. Values above 0.50 indicate good clustering; values above 0.70 indicate strong clustering. The Within-Cluster Sum of Squares (WCSS or Inertia) is used during the Elbow Method to determine optimal K. The Davies-Bouldin Index can optionally be computed as a secondary metric, where lower values indicate better-defined clusters. The Calinski-Harabasz Score (Variance Ratio Criterion) can also be used, where higher values indicate denser, more well-separated clusters.

Plots

The Correlation Heatmap visualizes pairwise Pearson correlations among all 20 features to detect redundancy and guide feature selection. Feature Distribution Histograms display the frequency distributions of each of the 20 features, revealing skewness, modality, and outlier presence. Box Plots by feature show the spread, median, and outliers for each variable across the dataset. The PCA Scatter Plot (pre-clustering) projects all regions into 2D PCA space to visually confirm whether natural groupings exist before K-Means is applied. The Elbow Method Plot shows WCSS on the y-axis versus K on the x-axis, with the elbow point highlighted as the optimal K. The Silhouette Score vs. K Plot shows silhouette values for each tested K to triangulate the optimal cluster count alongside the Elbow Method. The K-Means Cluster Visualization (post-clustering) is a 2D PCA scatter plot with each point colored by cluster label and centroid positions marked in a contrasting color

- **Required topics from the subject:**

- **Unsupervised Machine Learning** The core of the project is K-Means clustering, an unsupervised learning algorithm that partitions n observations into K clusters where each observation belongs to the cluster with the nearest centroid. Understanding the algorithmic mechanics — random initialization, iterative assignment, centroid update, and convergence — is essential. The concept of k-means++ initialization for better convergence must be covered.
- **Exploratory Data Analysis** Students must understand how to profile a dataset statistically and visually before modeling. This includes computing descriptive statistics, understanding skewness and kurtosis, detecting outliers using IQR, and interpreting correlation matrices.
- **Data Preprocessing and Feature Engineering** Topics include imputation strategies for missing values, the importance of feature scaling for distance-based algorithms, and the distinction between normalization and standardization. The impact of unscaled features on Euclidean distance-based clustering must be explained.
- **Dimensionality Reduction** PCA theory is required — understanding eigenvalues, eigenvectors, explained variance ratio, and the scree plot. The difference between PCA (linear) and t-SNE (non-linear) for visualization and their appropriate use cases must be understood.

- **Platform Required:**
- jupyter notebook
- **Books and Link Sources:**

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

<https://scikit-learn.org/stable/modules/clustering.html>

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html

https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.htm